**13** Write $x^2 + 6x - 7$ in the form $(x + a)^2 + b$ where $a$ and $b$ are integers.

| × | $x$ | 3 |
|---|---|---|
| $x$ | $x^2$ | $3x$ |
| 3 | $3x$ | 9 |

$(x + 3)^2 = x^2 + 6x + 9$

**1 mark**
Half of 6 is 3: form the square

$x^2 + 6x - 7 = (x + 3)^2 - 9 - 7$

$= (x + 3)^2 - 16$

**1 mark**
Subtract 16 to keep the expression equal

$(x + 3)^2 - 16$

**(Total for Question 13 is 2 marks)**

**14** Cone **A** and cone **B** are mathematically similar.
The ratio of the volume of cone **A** to the volume of cone **B** is 27 : 8

The surface area of cone **A** is 297 cm$^2$

Show that the surface area of cone **B** is 132 cm$^2$

**(Total for Question 14 is 3 marks)**

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